

习题二

$$2.4 (3) \quad F = A[\bar{B} + (C\bar{D} + EF)G]$$

$$\begin{aligned}\bar{F} &= \overline{A[\bar{B} + (C\bar{D} + EF)G]} \\ &= \bar{A} + B[(\bar{C} + D)(E + \bar{F}) + \bar{G}]\end{aligned}$$

$$F' = A + \bar{B}[(C + \bar{D})(\bar{E} + F) + G]$$

$$(4) \quad F = A + \overline{B + \overline{C + D + \bar{E}}}$$

$$\begin{aligned}\bar{F} &= \overline{A + \overline{B + \overline{C + D + \bar{E}}}} \\ &= \bar{A} \cdot \overline{\bar{B} \cdot C \cdot \bar{D} \cdot E}\end{aligned}$$

$$F' = A \cdot \overline{B \cdot \bar{C} \cdot D \cdot \bar{E}}$$

$$2.5 (1) \quad A = B$$

$$\Leftrightarrow A \cdot A = A \cdot B$$

$$\Leftrightarrow A = AB$$

$$\Leftrightarrow AB = A$$

正确

(2) $AB = AC$

当 $A=0$ 时

B, C 可为任意值

所以该逻辑关系不正确

(3) $A+B = A+C$

当 $A=1$ 时

B, C 可为任意值

所以该逻辑关系不正确

(4) 当 $A=1$ 时

$A+B = A+C$ 显然成立

又 $AB = AC$

则 $B=C$

当 $A=0$ 时

$AB = AC$ 显然成立

又 $A+B = A+C$

则 $B=C$

综上 $B=C$

2.b (3) $F = AB + AD + \bar{B}\bar{D} + A\bar{C}\bar{D}$
 $= A \cdot (B+D) + \bar{B}\bar{D} + A\bar{C}\bar{D}$

$$= A \cdot \overline{B} \overline{D} + \overline{B} \overline{D} + A \overline{C} \overline{D}$$

$$= A + \overline{B} \overline{D} + A \overline{C} \overline{D}$$

$$= A \cdot (1 + \overline{C} \overline{D}) + \overline{B} \overline{D}$$

$$= A + \overline{B} \overline{D}$$

$$2.7 \text{ c)} \quad F(A, B, C) = \overline{A} \overline{B} \cdot \overline{A} \overline{C}$$

$$= (A + \overline{B})(A + \overline{C})$$

$$= (A + \overline{B} + (C \cdot \overline{C}))(A + (\overline{B} \cdot \overline{B}) + \overline{C})$$

$$= (A + \overline{B} + C)(A + \overline{B} + \overline{C})(A + \overline{B} + \overline{C})$$

$$= \sum M(1, 2, 3)$$

$$F(A, B, C) = \sum m(0, 4, 5, 6, 7)$$

2.8 (1)

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

$$F = \sum m(0, 1, 2, 4)$$

2.10(2)

卡诺图

CD \ AB	00	01	11	10
00	1	1	1	0
01	1	1	1	0
11	1	0	1	1
10	1	0	1	1

$$F(A, B, C, D) = \bar{A}\bar{C} + AB + \bar{B}C$$

$$\bar{F}(A, B, C, D) = A\bar{B}\bar{C} + \bar{A}BC$$

$$F(A, B, C, D) = (\bar{A} + B + C)(A + \bar{B} + \bar{C})$$

2.11 F

CD \ AB	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

G

CD \ AB	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$$F(A, B, C, D) = \bar{D}$$

$$G(A, B, C, D) = D$$

故 $F = \bar{G}$

2.12 (1) 当 $a=1$ $b=0$ 时

$$F(A, B, C, D) = C\bar{D} + \bar{B}\bar{C} + A\bar{C}D$$

当 $a=0$ $b=1$ 时

$$F(A, B, C, D) = A\bar{C} + \bar{B}\bar{C} + \bar{A}C\bar{D} + ABC$$

故当 a 取 1 时能得到最简与或式。

(2) 当 $a=1$ $b=1$

$$F(A, B, C, D) = C\bar{D} + \bar{B}\bar{C} + A\bar{C}$$

2.13 (1)

AB \ CD	00	01	11	10
00	1	d	0	d
01	d	d	1	0
11	d	1	1	0
10	1	d	0	d

$$F(A, B, C, D) = \bar{A}\bar{B} + BD$$

(2)

AB \ CD	00	01	11	10
00	1	1	0	1
01	0	0	1	0
11	0	1	1	0
10	1	0	0	1

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	1	0	0
11	0	1	0	0
10	1	1	0	1

AB \ CD	00	01	11	10
00	0	1	0	0
01	0	0	0	0
11	1	1	0	0
10	1	0	0	0

$$F_1(A, B, C, D) = \bar{B}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD + ABD$$

$$F_2(A, B, C, D) = \bar{B}\bar{D} + \bar{A}\bar{C}D + \bar{A}BC$$

$$F_3(A, B, C, D) = \bar{A}\bar{B}C + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD$$